

Pedro Silva

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EDUCATION

Instituto Superior Técnico, University of Lisbon

MSc in Computer Science and Engineering

Lisbon, Portugal

Sep. 2025 – Aug. 2027 (expected)

Faculty of Sciences, University of Lisbon (FCUL)

BSc in Informatics Engineering

Lisbon, Portugal

Sep. 2022 – Aug. 2025

EXPERIENCE

Software & AI Team Member

2025 – Present

AeroTec ATLAS, Systems & Aeronautics (IST UAV team)

Lisbon, Portugal

- Develop computer vision and autonomy software for the only Portuguese university team flying fully autonomous drones
- Built Atlas Pipeline, a modular Python library for real-time video capture, processing and streaming (GStreamer) with ROS 2 integration and NVIDIA DeepStream AI inference for object detection
- Developed a ground-station camera system to command vehicles (minidrone and VTOL), receive live video feeds and monitor device status, with automated dev-environment setup scripts

Programming Content Creator

2021 – Present

YouTube channel zFake

youtube.com/c/zFake

- Teach programming and computer science to a Portuguese-speaking audience, including a 2D game development series in vanilla JavaScript with open-source companion code

Avionics Team Member, PalmaSat

2022

CanSat Portugal, 9th edition (ESERO)

Portugal

- Co-built the flight software of a can-sized microsatellite (telemetry, sensors, radio downlink); awarded the national **Technical Performance Award**

PROJECTS

Sophia | TypeScript, Node.js, discord.js, LLM tools, SQLite | github.com/zF4ke/sophia

- Built a production-grade conversational AI agent platform for Discord with 33 tools (moderation, role management, server search) driven by native LLM function calling in an agentic loop
- Designed the safety model that makes it production-ready: every server change needs admin approval, destructive actions require a second confirmation and can never be auto-approved, all under role-based access and per-conversation budget limits

Timebox | TypeScript, Electron, multi-agent LLM systems, Vitest | github.com/zF4ke/timebox

- Built a desktop app where five specialist LLM agents draft, critique and vote on a weekly schedule until they reach consensus, then export it straight to your calendar
- Built a benchmarking harness that compares models on schedule quality versus cost, with hard budget caps and a deterministic scorer that flags real mistakes like missed deadlines and late-night work

Traveller | TypeScript monorepo, Next.js, MongoDB, discord.js

- Created a multiplayer RPG played inside Discord where decrypting ciphers drives the story, built on a platform-agnostic engine whose test suite plays entire quests in under a second

ModernBazaar | Java, Spring Boot, TypeScript, Next.js, PostgreSQL | github.com/zF4ke/ModernBazaar

- Built a market analyzer for a live game economy: a Spring Boot backend ingests and compacts price data while a Next.js dashboard shows real-time analytics and historical trends

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Java, C/C++, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, Spring Boot, Tailwind CSS, ROS 2

AI & Robotics: PyTorch, OpenCV, GStreamer, NVIDIA DeepStream, LLM agents and function calling, neural networks

Developer Tools: Git, Linux, Docker, PostgreSQL, MongoDB, Firebase

Courses: Neural Networks: Zero to Hero (Andrej Karpathy), Node Basic (HackerRank), NLW IA (Rocketseat)

Spoken Languages: Portuguese (native), English (fluent)